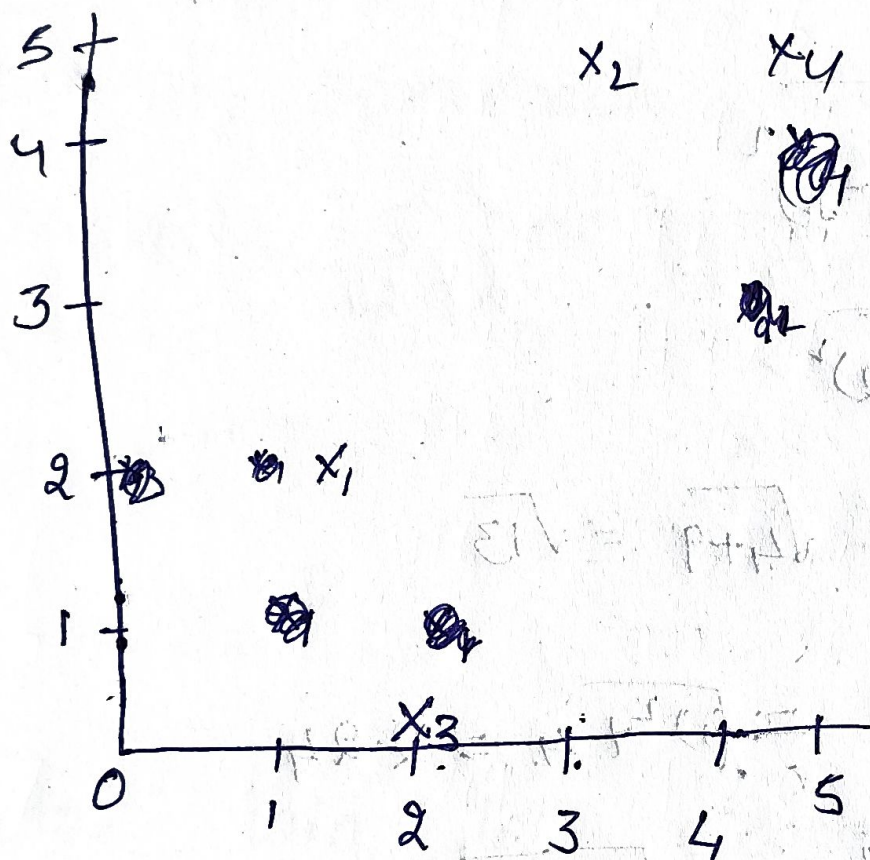


Activity-5

Minkowski distance

Consider the following $(1, 2), (3, 5), (2, 0), (4, 5)$ Consider the data matrix and similarity matrix by Manhattan & Supremum measure



Point	attribute	attribute
x_1	1	2
x_2	3	5
x_3	2	0
x_4	4	5

Similarity matrix using Manhattan method

	x_1	x_2	x_3	x_4	x_5
x_1	0	X	X	X	
x_2	5	0	X	X	
x_3	3	6	0	X	
x_4	6	1	7	0	

x_1	0	X	X	X
x_2	3.61	0	X	X
x_3	2.24	5.1	0	
x_4	4.24	1	5.39	0

$$x_2 = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$$= \sqrt{(2-1)^2 + (5-2)^2}$$

$$= \sqrt{(2)^2 + (3)^2} = \sqrt{4+9} = \sqrt{13}$$

$$x_3 = \sqrt{(2-1)^2 + (-2)^2} = \sqrt{(1)^2 + 4} = 2.24$$

$$x_3 = \sqrt{(2-3)^2 + (-5)^2} = \sqrt{(-1)^2 + 25} = 5.1$$

$$x_4 = \sqrt{(3)^2 + (3)^2} = \sqrt{9+9} = 4.24$$

$$x_4 = \sqrt{1} = 1$$

$$x_4 = \sqrt{(2)^2 + (5)^2} = \sqrt{4+25} = \sqrt{29} = 5.39$$

Supplement

	x_1	x_2	x_3	x_4
x_1	0			
x_2	3	0		
x_3	2	5	0	
x_4	3	1	5	0

Document team coach hockey baseball soccer

Document 1 5 0 3 0 2

11 2 3 0 2 0 1

11 3 0 7 0 2 6

11 4 0 1 0 0 2

Score win loss season

6 2 0 0

0 1 0 1

0 3 0 0

2 0 3 0

$$\cos(d_1, d_2) = \frac{(d_1 \cdot d_2)}{||d_1|| ||d_2||}$$

$$d \cdot d_2 = 15 + 6 + 2 + 2 = 25$$

$$\|d_1\| = \sqrt{5^2 + 0^2 + 3^2 + 0^2 + 2^2 + 0^2 + 0^2 + 2^2 + 0^2 + 0^2}$$

$$= \sqrt{25 + 9 + 4 + 4} = \sqrt{42} = 6.480$$

$$\|d_2\| = \sqrt{3^2 + 2^2 + 1^2 + 1^2 + 1^2 + 1^2} = \sqrt{9 + 4 + 1 + 1 + 1 + 1} = \sqrt{17} = 4.123$$

$$\cos(d_1, d_2) = \frac{25}{6.480 \times 4.123} = \frac{25}{26.71704} = 0.935$$